PRAYATNA SAPKOTA

Geomatics Engineering Student

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PROFESSIONAL PROFILE

Motivated Geomatics Engineering undergraduate at Kathmandu University with a strong interest in astronomy, space science, and Earth observation systems. An independent and disciplined learner dedicated to building rigorous technical foundations in engineering, geospatial sciences, and scientific inquiry. Actively pursuing opportunities to contribute to student research, technical workshops, and collaborative academic programs.

EDUCATION

Kathmandu University
2026 – Present

Bachelor of Engineering (B.E.) in Geomatics Engineering (2026 Batch).

Developing foundational knowledge in spatial analysis, surveying engineering, mathematics, and physical sciences.

Banepa NIST Secondary School
Higher Secondary (+2)

Higher Secondary Education

Completed secondary academic coursework with a core emphasis on science and mathematics.

RESEARCH & INDEPENDENT WORK

A Conceptual Hypothesis for Microbial Survival on Rogue Planets
2026

Independent Conceptual Astrobiology Research

- Authored a conceptual hypothesis examining theoretical mechanisms for microbial dormancy, planetary ejection, and long-duration interstellar transport on unbound planetary bodies.
- Formulated a structured qualitative framework linking astrobiological preservation models with interstellar drift environment parameters.
- Published independently on Zenodo repository platform: https://zenodo.org/records/18859992

SKILLS & COMPETENCIES

Computer Literacy:	General computer operation, digital document processing, web navigation, research archiving.
Learning & Research:	Independent scientific study, literature review, conceptual hypothesis formulation, self-directed learning.
Academic Foundations:	Geomatics engineering fundamentals, spatial reasoning, introductory engineering mathematics.

ACADEMIC & SCIENTIFIC INTERESTS

- **Astronomy & Space Science:** Astrobiology, planetary systems, rogue planet dynamics, and celestial mechanics.
- **Geomatics Engineering:** Satellite positioning, Earth observation, remote sensing principles, and spatial data systems.
- **Independent Scientific Inquiry:** Interdisciplinary research, physics concepts, and scientific methodology.